

# Influence of indoor soundscape perception based on audiovisual contents on work-related quality with preference and perceived productivity in open-plan offices

Hyun In Jo, Jin Yong Jeon<sup>\*</sup>

Department of Architectural Engineering, Hanyang University, Seoul, 04763, South Korea

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## ABSTRACT

Open-plan offices (OPOs) are being increasingly implemented to promote communication and collaboration among coworkers. We investigated the effects of indoor soundscape perception on productivity and preferences for the work environment in OPOs containing various audiovisual contents to achieve high work-related quality. A subjective evaluation was performed through two experiments conducted in a virtual reality environment with various stimuli. In Experiment 1, we investigated the correlations of visual complexity and acoustical variability with work-related quality. The results on a seven-point Likert-like scale (−3 to 3) indicated that visual complexity and maximum appropriate acoustical variability of an OPO was recommended at −1.3 to −0.2 and −0.3, respectively. In Experiment 2, the participants identified various noise sources and responded on their effects on the perceived affective quality and work-related quality. The results revealed that indoor soundscape perception can be interpreted in a comfort–content dimension and is significantly influenced by worker interactions such as conversation and laughter. Increased comfort improved the work environment preferences, and increased content significantly influenced the perceived productivity. Considering the factor score of the comfort–content dimensions, these results suggest the need to design a balanced OPO sound environment with preferred and productive zones by ensuring comfort and content distributions in the ranges of 0.3–0.7 and 0.6–1.2, respectively. The need to achieve an appropriate level of content in the indoor soundscape is confirmed, contradicting the notion that OPOs should be quiet. Therefore, these results can guide the design of sound environments in open office spaces.

## 1. Introduction

Numerous companies have set up open-plan offices (OPOs) owing to increasing awareness of the importance of communication and collaboration among coworkers in offices. Therefore, researchers have investigated methods to improve various indoor environmental quality factors (e.g., layout, air quality, thermal comfort, lighting, and acoustic comfort) to achieve optimal worker productivity and satisfaction in OPOs [1–3]. Among these factors, noise has been confirmed as the most critical factor influencing the productivity and satisfaction of office workers [4,5]. Given that OPOs are vulnerable to noise exposure unlike conventional enclosed-type offices, a poor office noise environment has a negative influence on work satisfaction and task performance in OPOs [6–9]. Thus, the understanding and management of the complex sound environment is indispensable given that OPOs require the simultaneous

consideration of co-occupancy management, communication, and focus for personal work activities [10].

Sound pressure level and sound privacy are the major factors influencing the acoustic quality of the OPO sound environment [1,11,12]. When the noise level is increased, workers may experience increased fatigue and a decrease in their cognitive functions. Therefore, it is critical to maintain a quiet sound environment in an OPO [13–16]. Furthermore, the negative influence of such office noise can vary depending on the type of noise. Among the various noise sources in an office (e.g., communication, telephone, and speech noise), conversation noise, which is referred to as irrelevant speech noise, is a primary factor that causes annoyance [17–19]. In addition, speech-related noise (irrelevant speech and intelligible conversation noise) causes psychological dissatisfaction, lowers work productivity, and increases workload [20–23]. Therefore, speech privacy should be considered to

<sup>\*</sup> Corresponding author.

E-mail address: [jjyeon@hanyang.ac.kr](mailto:jjyeon@hanyang.ac.kr) (J.Y. Jeon).

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increase the work satisfaction and work-related quality of workers in OPOs [12,24–26]. In addition to speech-related noise, telephone noise and fan noise have negative influences on the performance and satisfaction of workers [6,27]. Thus, attempts are being made to establish design standards for OPO sound environments. The ISO 3382-3 standard [28] suggests target values of appropriate levels based on various parameters such as A-weighted sound pressure level (SPL) at 4 m, spatial decay rate, and distraction distance. Thus, there have been attempts to utilize various building interior elements such as workstations, partitions, and office layouts to ensure speech privacy in OPOs [14,29,30]. Furthermore, the recently established ISO 22955 standard classifies six types of OPOs and suggests activities that can occur in each type of OPO, in addition to acoustic design standards in accordance with these activities.

In summary, previous studies conducted on the OPO sound environment focused on various noise sources that occur in most offices with respect to waste, as well as on the reduction of the background noise level or use of sound-absorbing elements to achieve a quiet office environment. However, the annoyance caused by noise in offices cannot be determined based on the SPL, which is expressed as  $L_{Aeq}$  [23], and the definition of noise based on specific types or sound magnitudes remains limited [17]. Therefore, individual responses to a sound environment composed of various environmental factors needs to be considered because the human perception of the OPO sound environment is yet to be understood based on physical parameters [31,32].

Given this context, the concept of a soundscape, which interprets the sound environment with respect to human perception instead of focusing on the physical noise level while considering sound as a resource, was developed [33]. The ISO 12913-1 standard [34] defines soundscape as an “acoustic environment perceived or experienced and/or understood by a person or people, in context.” Auditory sensation and interpretation may vary depending on changes in context. Consequently, context is a significant influencing factor for the perception of the soundscape [35,36]. Various studies performed soundscape evaluation and investigated analysis methods for outdoor environments [32,37]; this led to the establishment of the ISO 12913-2 [38] and ISO 12913-3 [39] standards. For an outdoor soundscape, the representative perceptual dimensions of pleasantness and eventfulness have been standardized, which enable mutual comparisons of research results obtained by different researchers [38,40]. These studies indicate that there has been an increase in the research on the evaluation of soundscapes for external spaces with various functions (e.g., urban, rural, and parks), and the perception model is well established [41–44]. In contrast, soundscape evaluation studies on indoor environments are limited [45–47]. Auditory perception varies across various contexts such as space functions, finishing materials, and spatial characteristics, which are characteristics that are more conspicuous in indoor environments than in outdoor environments [48]. Therefore, soundscape research on indoor environments is indispensable. Initial attempts were made to analyze soundscape recognition for various indoor environments such as residential buildings, historical spaces, public spaces, and libraries based on the function of the environment [45,47,49]. OPOs require extensive research because of the various types of internal sound sources and multiple construction planning factors. Consequently, one study investigated the soundscape of OPOs, and a perceptual framework of an indoor soundscape was proposed using qualitative research based on interviews [50]. However, a limited number of studies have investigated the indoor soundscape of OPOs. Comfort, content, and familiarity have been recently suggested as perceptual dimensions for analyzing indoor soundscapes [45]. In other words, the existing studies [45,50] present only a conceptual framework or evaluation metrics for interpreting the indoor soundscape of the OPOs. Thus, if the soundscape perceptual dimension proposed by a previous study [45] is applied to an OPO, it will be possible to obtain an in-depth understanding of the indoor soundscape and suggest design methods to implement a comfortable OPO indoor soundscape through interpretation based on quantitative

research.

When analyzing soundscapes, it is critical to consider multisensory perceptions. Among these perceptions, audiovisual interactions have been researched actively [41,51,52]. Spaces shared by multiple people, such as OPOs, require a more detailed examination of the influences of audiovisual interactions. Hence, research was conducted to investigate the influence of audiovisual interactions on the soundscape design of indoor environments [53]. Most studies focused on the positive effects of visual natural stimuli on the reduction of annoyance caused by noises (such as aircraft noise and traffic noise) in a human-inhabited environment. In addition to such changes in the indoor context, changes in the perception of living noises (such as floor impact noise and traffic noise) were investigated based on whether audiovisual information was provided [54–56]. The results revealed the annoyance reaction to noise changes based on whether visual information was provided. In a previous study on audiovisual interaction for OPOs, changes in preferences were observed depending on whether audio and visual information on various water features was provided, and significant differences in the preferences were observed based on the installation location of the water features [57]. Given that OPOs contain various visual contexts based on the interiors and layouts, the effects of these visual contexts should be considered when analyzing indoor soundscapes. However, few studies have considered audiovisual content in the analysis of the indoor soundscape of OPOs.

In summary, attempts are being made to propose guidelines for sound environment design by applying the soundscape concept to an indoor environment, and to consider non-acoustic factors such as vision in addition to acoustic factors in the context of the building interior. However, early studies were conducted on indoor spaces such as residential spaces, libraries, and religious facilities, but few studies have been conducted on OPOs. Hence, this study can be said to be novel in that it applies the indoor soundscape concept to the OPO sound environment. In contrast to previous studies, this study proposes an indoor soundscape perception model based on the workers’ actual experience and perception of the sound environment considering various audiovisual contents in the OPOs from the perspective of a soundscape. This study proposes a strategy for improving the OPO sound environment such that it can achieve work environment satisfaction and work productivity. The research questions for this study are as follows:

- 1) How does work-related quality change in accordance with changes in audiovisual content?
- 2) What causes correlations between the indoor soundscape perceptual dimension and work-related quality?
- 3) How should the OPO indoor soundscape be designed to improve work performance and preference for the work environment?

The remainder of this paper is organized as follows. Section 2 describes the experiment conducted to investigate the correlation between audiovisual content and work-related quality and presents the results. Section 3 describes the experiment conducted to determine the correlation between the indoor soundscape perception and work-related quality according to various changes in the OPO acoustic content and presents the results obtained. Section 4 discusses the results, explains the limitations of the study, and presents the future scope. Finally, Section 5 concludes the paper.

## 2. Experiment 1: audiovisual content and work-related quality

### 2.1. Methods

#### 2.1.1. Sound stimuli

The correlation between audiovisual content and work-related quality was investigated in Experiment 1. The sound stimuli used in the subjective evaluation were generated based on computer simulations [58]. Field measurements were performed in an OPO in downtown

Seoul to collect the information required for the computer simulations. The selected OPO had dimensions of 20.2 m (length)  $\times$  15.2 m (depth). The ceilings and floors were finished with absorption gypsum boards and vinyl tiles, and 1.1 m partitions were installed inside. Acoustic measurements were conducted in accordance with the ISO 3382-3 standard [28]. An omnidirectional speaker (Type 4292-L, Brüel & Kjær) and omnidirectional half-inch microphone (GRAS AE 46, GRAS Sound and Vibration) were used for these measurements. Omnidirectional sound sources that match the sound power spectrum of the general voice suggested in the ANSI S 3.5–1997 (R 2007) standard were used, and seven measurement points were selected along the lines across the unit work spaces. For the recorded sound sources, three single-number quantities ( $L_{p,A,S,4m}$ ,  $D_{2,S}$ , and  $r_D$ ) suggested in the ISO 3382-3 standard [28] were analyzed. Here,  $L_{p,A,S,4m}$ ,  $D_{2,S}$  (spatial decay rate of speech), and  $r_D$  (distraction distance) denote the A-weighted SPL at a distance of 4 m from the sound source, decay rate of the A-weighted SPL as the distance doubles, and distance at which the speech transmission index from the speaker decreases below 0.5, respectively. The calculation results for  $L_{p,A,S,4m}$ ,  $D_{2,S}$ , and  $r_D$  were 49.4 dB, 4.6 dB, and 11.1 m, respectively. The computer simulation model was implemented based on actual field measurements using the Odeon Simulation (version 15) software application. The walls, ceilings, floors, desks, and drawers inside the office were modeled by applying sound absorption coefficients similar to those of actual materials. Consequently, the  $L_{p,A,S,4m}$ ,  $D_{2,S}$ , and  $r_D$  values were found to be highly similar to those of actual office spaces within a 2% error range. Hence, the computer simulation model in this study was considered to be well implemented and similar to the actual case.

Next, three acoustic models were generated by varying the sound absorption coefficients of the ceiling and floor in the simulation. Given

that  $L_{p,A,S,4m}$  was within the range of 44.3–53.8 dB in a previous case study on OPOs [59], the  $L_{p,A,S,4m}$  values were set as 45.9 dB (low), 51.1 dB (middle), and 55.9 dB (high) to include the general range, as shown in Fig. 1, with reference to the previous study [58]. Consequently, three-room impulse responses were generated. The generated room impulse responses and speech sound sources with durations of 1 min, as recorded in an anechoic room, were then convolved. This sound source has a high SPL for the frequency band below 500 Hz, similar to the characteristic of the speech sound source in the ISO 3382-3 standard [28]. The SPLs of the convolved sound source were adjusted to be identical to the three abovementioned  $L_{p,A,S,4m}$  values.

The speech sound sources generated through convolution did not include the background noise of the office because anechoic room sound sources were used. Hence, the background noise was recorded separately in an actual OPO. This office was used for business, had dimensions of 25.0 m (length)  $\times$  14.5 m (depth), and the ceiling and floor were finished with an acoustic board and carpet. Measurements were performed for 1 h using a sound level meter (SVAN 953, Svantek, Warsaw, Poland) while workers occupied the internal space during the daytime working hours on a weekday. The measurement height was 1.2 m, which is the average ear height of workers in the sitting position. A background noise source with duration of 1 min was separately edited in a section where there was no worker speech in the recorded background noise source; the background noise  $L_{Aeq}$  was 47.0 dB. The sound source length for the evaluation stimuli was set as 1 min with reference to a previous study [60,61] wherein the perception of the office sound environment was evaluated using sound source stimuli with duration of less than 1 min in a laboratory environment. Thus, three sound stimuli were secured by synthesizing sound sources produced through the convolution and the edited background noise source. The signal-to-noise

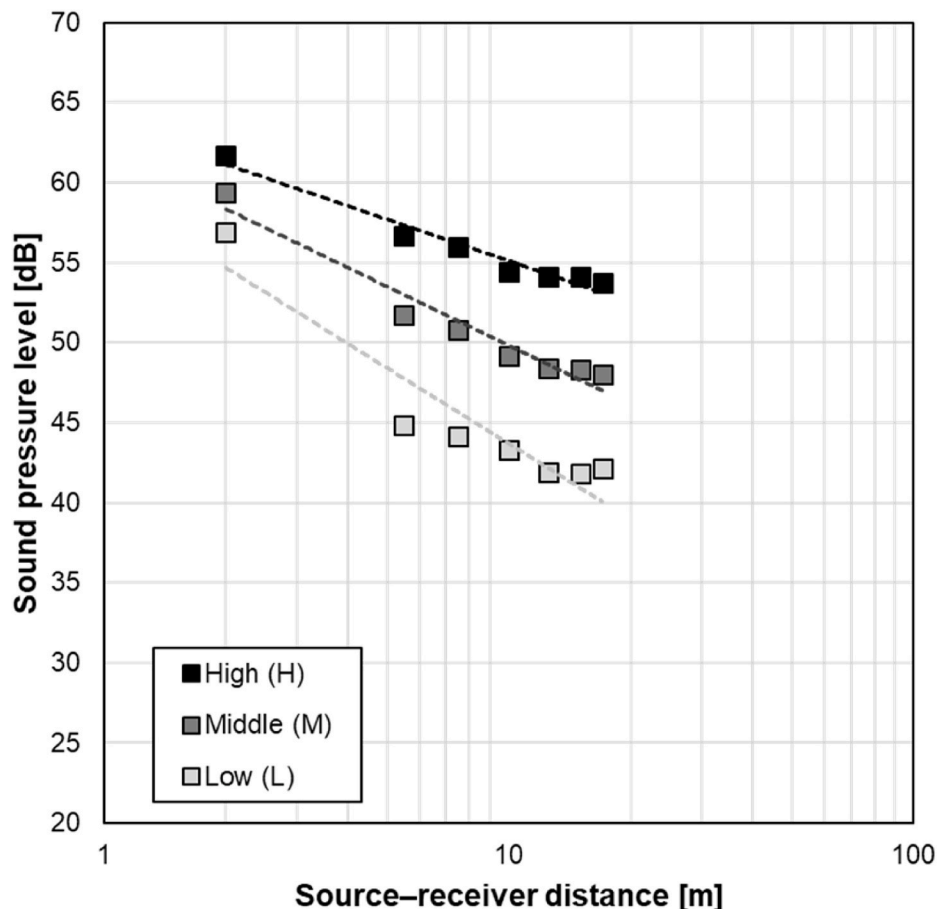


Fig. 1. Sound pressure level of simulated open-plan office according to source–receiver distance.

ratios of each sound source stimulus were  $-1.1$  dB (low),  $4.1$  dB (middle), and  $8.9$  dB (high).

### 2.1.2. Visual stimuli

Various visual data were obtained using virtual reality (VR) technology to provide a sense of immersion and reality similar to an actual OPO environment when performing the subjective evaluation of sound source stimuli in a limited laboratory environment. The VR office environment was implemented using Unity 3D software, which is a cross-platform game engine. Four OPO spaces with different sizes, internal layouts, interiors, and capacities were implemented to include sufficient variations for the visual environment of a general OPO, as shown in Fig. 2. Office 1 was the smallest office space with a floor area of  $57\text{ m}^2$ , and it could accommodate approximately five workers in a small office plan that only included a working space. Office 2 had an office plan with a floor area of approximately  $160\text{ m}^2$  that could accommodate 15 workers and included working and resting spaces. Office 3 had a double-floor structure with a height of  $7.3\text{ m}$  and a floor area of  $550\text{ m}^2$ ; it could accommodate 32 workers. It had a relatively large office plan that included working, resting, and meeting spaces. Office 4 had a single floor structure with a floor area of  $484\text{ m}^2$ , and the same height as that of Office 3. Its office plan included working and resting spaces. The ratios of the window area to the total wall area of the four office models ranged from 10% to 45%. There were two types of sceneries outside the windows of the OPOs: an urban background with orderly infrastructure and a backward urban environment. The interior of every office was maintained in a clean condition to control factors besides the visual factors of the building design such as the interior, partitions, and furniture.

The evaluation positions in the VR offices were selected, and the same direction was specified for all participants, who were instructed to face adjacent windows at the start of the evaluation. The view height at these positions was fixed at  $1.2\text{ m}$ , which was the same as the sound stimuli measurement height. However, after the start of the evaluation, participants were allowed to adapt to the environment by rotating their heads freely to view the environment. Consequently, a total of eight visual stimuli were obtained (four office types  $\times$  two window views = eight visual stimuli).

### 2.1.3. Questionnaire and work performance quantification task

The questionnaire used for the participant evaluation in Experiment 1 consisted of two parts:

- 1) Questions about audiovisual content in the OPO: The participants were instructed to evaluate the complexity of the visual environment

of the given office and the variability of the auditory environment by providing responses on a seven-point Likert-like scale ( $-3$ : *not at all* to  $3$ : *extremely*).

- 2) Questions about work-related quality: The participants provided responses on their overall satisfaction with the office work environment on a seven-point Likert-like scale ( $-3$ : *not at all* to  $3$ : *extremely*) from a psychological aspect.

In addition, the participants performed a backward digit span task in each stimulus environment to quantify the work performance from a physical aspect [58]. The backward digit span task was selected to evaluate the short-term verbal memory [62]. In this task, random number arrays were provided to the participants through a separate speaker, and the participants were requested to verbalize the given number array in reverse. Each array initially contained a minimum of two numbers, and one number was added after a participant provided two successive correct responses. The number of arrays was increased to eight. If a participant provided the correct response to all questions in the task, they could perform the task for a total of 16 questions. The tester recorded the verbal responses of the participants, and the ratios of the correct responses to all questions were calculated as a percentage to represent the task performance.

### 2.1.4. Procedure

G-power software (Version 3.1.9.4) was used to select an appropriate sample size. A minimum of 31 participants were required to perform the verification with an effect size of 0.35 and a verification power of 0.8. Thus, 34 participants were recruited for this study (25 males and 9 females); their ages ranged from 20 to 33 years old (mean: 24.85 and standard deviation: 3.56). Individuals who had been attending the same university and had experience of working in an office space were recruited to reduce the deviations in the responses of the participants. Before starting the experiment, written consent for participation in this study was obtained from every subject as an ethical procedure. Furthermore, they were provided with a full explanation regarding the objective of the experiment and the content of the questionnaire. In addition, the participants were instructed on how to use the VR devices and were allowed to sufficiently familiarize themselves with the devices. The evaluation environment was constructed in a quiet laboratory. The walls were finished with porous sound-absorbing panels, and the background noise was low (approximately 30 dBA). The temperature in the laboratory environment was maintained at  $23\text{--}24\text{ }^{\circ}\text{C}$ , which was suggested as an appropriate temperature for an office space in previous studies [63,64]. A total of 24 stimuli (3 audio  $\times$  8 visual stimuli) were

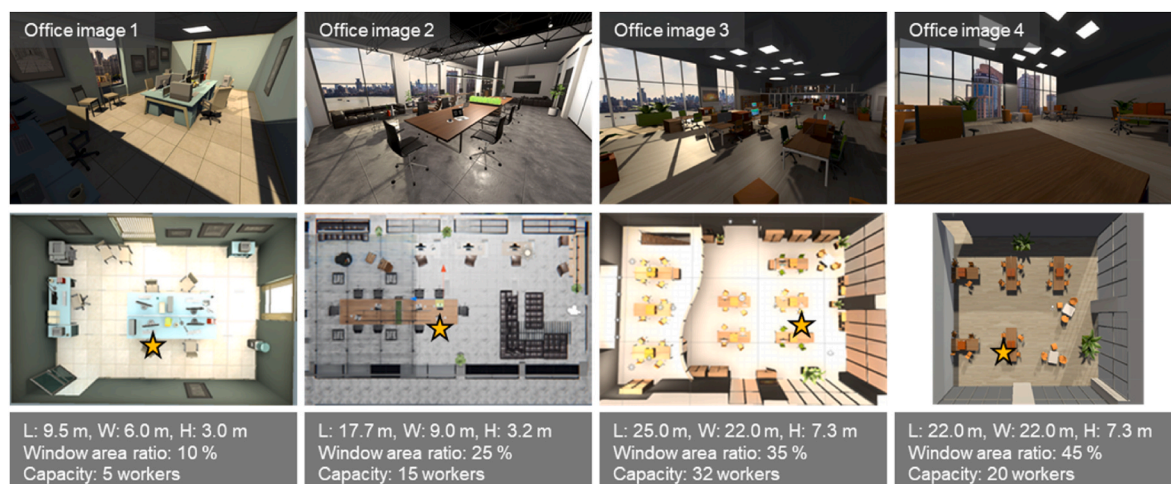
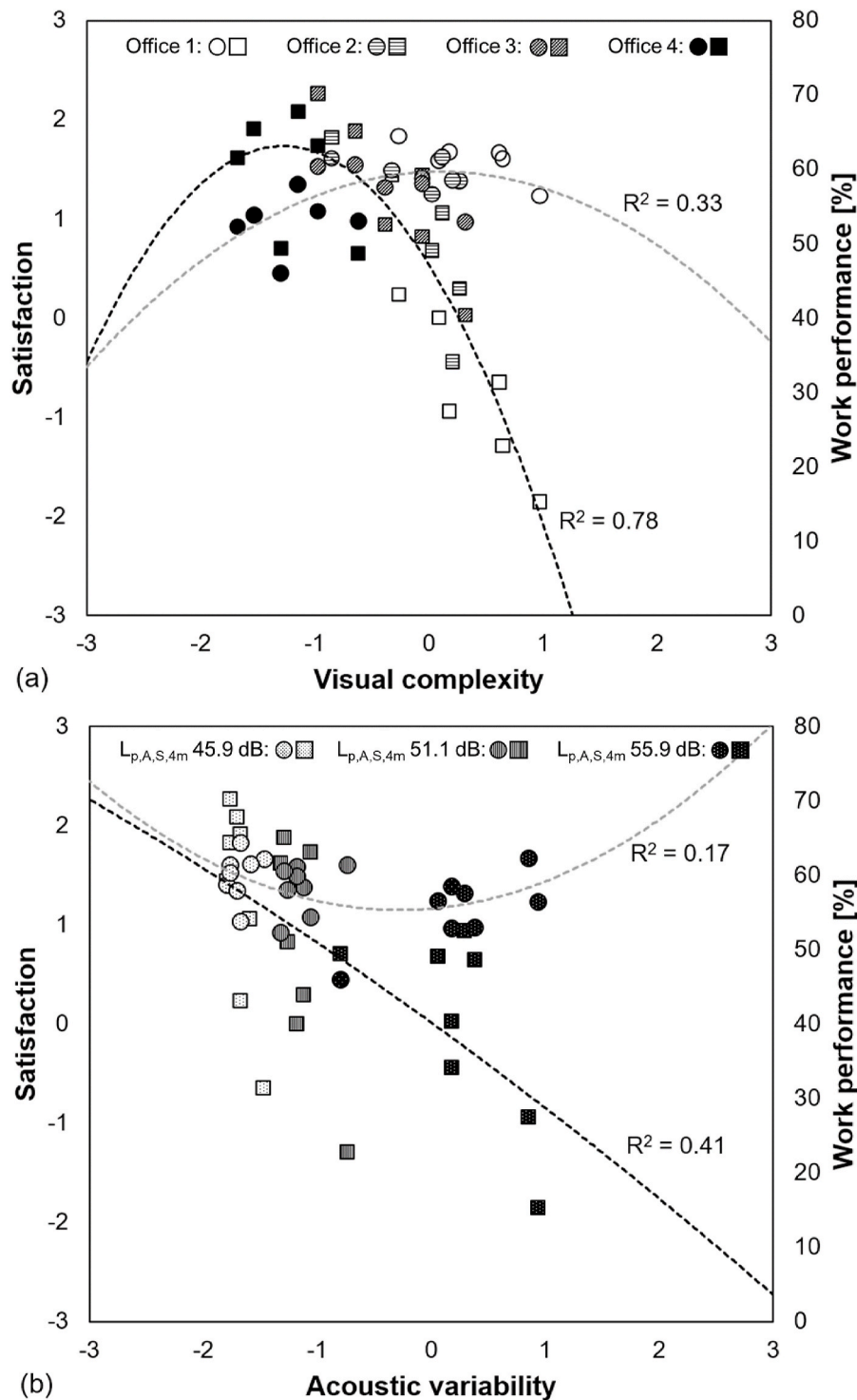


Fig. 2. Physical characteristics and interior views of four VR OPO visual stimuli (yellow stars indicate the experiment positions). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)



provided randomly by varying the combinations of the audiovisual stimuli. The sound stimuli were provided through an open-type headphone (Sennheiser HD 650), and the visual stimuli were provided through a head-mounted display (VIVE Pro EYE). The subjects experienced each evaluation stimulus for 1 min before responding to the questionnaire. The stimuli were provided repeatedly according to the subjects' requests. Every question was provided through a graphical user interface (GUI) in a VR environment. After the questionnaire was completed, the participants performed the backward digit span task.

After the evaluation for one stimulus was completed, silence was maintained for 10 s before the next stimulus, and this process was repeated. The total duration of the experiment was limited to less than an hour considering the fatigue of subjects and adverse reactions (e.g., dizziness, nausea) from wearing the VR equipment for a long time. A total of 816 responses ( $24 \text{ stimuli} \times 34 \text{ participants} = 816$ ) were obtained for each evaluation item [58].



**Fig. 3.** Work-related quality for different audiovisual contents in an OPO: (a) visual complexity and (b) acoustic variability (squares denote the satisfaction and circles denote the work performance).

## 2.2. Results

Responses related to the work-related quality according to the variation in the audiovisual contents of the OPO were segregated into visual complexity and acoustical variability, as shown in Fig. 3. In this figure, the average responses related to work satisfaction and work performance for the 24 stimuli are indicated. To verify the generalized response trend of the work-related quality according to the changes in the visual complexity and acoustical variability in the OPO, second order polynomial regression analysis was conducted, and the results are shown in Fig. 3.

First, as shown in Fig. 3 (a), the visual complexity is the highest in Office 1 (0.37), followed by Office 2 (−0.09), Office 3 (−0.30), and Office 4 (−1.21). With an increase in visual complexity, the work performance and satisfaction with the work environment decreased. Based on the seven-point Likert-like scale (−3 to 3), the satisfaction was the highest at 1.74 when the visual complexity was −1.28, and the work performance was the highest at 59.68% when the visual complexity was −0.10. With respect to the office type, the average satisfaction was the highest in Office 4 (1.45), followed by Office 3 (1.23), Office 2 (0.81), and Office 1 (−0.75); the average work performance was the highest in Office 1 (61.31%), followed by Office 2 (59.39%), Office 3 (58.08%), and Office 4 (52.89%), which demonstrates an opposite trend. The average satisfaction and work performance were higher in a high-class landscape with a well-arranged exterior view than in a poor landscape. Consequently, there was a difference of approximately 1.18 in the visual complexity levels between psychological satisfaction, which is required by priority to achieve maximum satisfaction from the perspective of the workers (employees), and physical work performance, which is required by priority to achieve maximum work performance from the perspective of the manager (employer).

Regarding the correlation between acoustic variability and work-related quality, the acoustic variability is the highest when  $L_{p,A,S,4m}$  is 45.9 dB (−1.68), followed by 51.1 dB (0.81) and 55.9 dB (−0.03), as shown in Fig. 3 (b). Based on the seven-point Likert-like scale (−3 to 3), work performance was the lowest at 55.35% when the acoustic variability was −0.28, and the satisfaction decreased as the variability increased. As the variability increased with the audio content of the sound environment, the satisfaction and work performance decreased to a certain extent. However, when the variability was higher than −0.28, the work productivity remained the same or increased slightly, whereas the work satisfaction decreased continuously. The satisfaction was the highest when  $L_{p,A,S,4m}$  was 45.9 dB (1.27), followed by 51.1 dB (0.81) and 55.9 dB (−0.03). Furthermore, the work performance was the highest at 45.9 dB (60.08%), followed by 51.1 dB (58.25%) and 55.9 dB (55.40%). However, the  $R^2$  of acoustic variability was lower than that of visual complexity, which failed to reach a statistical significance level. This was because the diversity of the audio content was not secured when compared with the visual environment, given that only the sound absorption coefficient was varied while using the same speech and background noise sources for the sound stimuli used in Experiment 1. Therefore, Experiment 2 was performed as an additional experiment to examine the positive effects of acoustic variability on work-related quality accurately while overcoming the limitations of Experiment 1 by sufficiently including various noise events that occur in actual office spaces.

## 3. Experiment 2: Indoor soundscape perception with various OPO acoustic contents

### 3.1. Methods

#### 3.1.1. Audiovisual stimuli

In Experiment 2, the correlation between the indoor soundscape perception and work-related quality according to various changes in the OPO acoustic contents was investigated by expanding Experiment 1.

Hence, a total of 12 sound stimuli were secured by editing sound source samples with durations of 1 min, such that they included various types of noise sources (e.g., conversation, computer, and footsteps) in the collected background noise sources, as presented in Section 2.1.1. The physical acoustic characteristics of each sound source were analyzed in terms of the sound strength, temporal variation, and spectral content, as shown in Table 1 and Fig. 4. First, the sound strength was quantified using an A-weighted SPL ( $L_{Aeq}$ ).  $L_{A10-A90}$ , which denotes the difference between the percentile SPLs  $L_{A10}$  and  $L_{A90}$ , was analyzed for the temporal variability. In addition, the SPL per 1/1 octave frequency band was analyzed for the spectral content. Thus, the  $L_{Aeq}$  of the selected 12 stimuli ranged from 49.9 to 59.3 dB, and  $L_{A10-A90}$  ranged from 2.7 to 12.6 dB. Furthermore, the range of the SPLs per frequency band, as shown in Fig. 4, indicates that the difference between the sound source samples increases as the frequency band shifts from a low band to a high band. The results revealed that the maximum and minimum values of each acoustic parameter exhibited a difference of approximately 10 dB, and the difference in the SPL between the frequency bands was 10 dB or higher at frequencies of 1 kHz or higher. This observation suggests that the 12 stimuli selected in this study sufficiently represent the general sound environment of OPOs.

The frequencies of the 12 types of noise sources for the 12 sound source stimuli were analyzed as shown in Table 2. Given that the conversation noise caused by communication among workers inside the office and the fan noise of the heating, ventilation, and air conditioning (HVAC) system, which is usually operational at all times, are difficult to represent using the number of occurrences, their durations were determined in seconds. Three researchers majoring in architectural acoustics listened to each sound source repeatedly and then determined and verified the number of occurrences to accurately represent the occurrence frequency per noise source.

The results revealed that noise events caused by mechanical fans and air conditioning (AC), conversation, and computer peripherals such as a keyboard and mouse occurred most frequently in the OPO sound environment selected in this study.

Walking and footstep noise caused by the movements of workers occurred frequently, i.e., more than 10 occurrences (#5, #8, #9, #11, and #12). Furthermore, chair dragging (#9, #10, and #12) and desk noise (#6) caused by the collisions of pens and other objects with the desks occurred frequently. Noise sources caused by work processes or personal activities of workers such as turning of pages, operation of printers and copiers, slamming of doors, water purifiers, clinking of glass cups, and laughter occurred infrequently, fewer than 10 times. Hence, the results of the physical acoustic parameters reconfirmed that the 12 sound source samples used in this study represented sufficiently diverse OPO sound environments.

As demonstrated in Experiment 1, it is critical to consider the visual and auditory environments when evaluating the indoor soundscape perception of OPOs. Therefore, among the office models used in Experiment 1, offices 1, 2, and 3 were selected and used as VR visual stimuli. Thus, only urban backgrounds with well-arranged infrastructure were selected for the exterior views of the VR offices. In addition, the same evaluation position and view height as those in Experiment 1 were selected. However, Office 4 was excluded considering the total experiment time, given that its size is similar to that of Office 3, and to minimize adverse reactions caused by the long-term wearing of the head-mounted display. As a result, 36 audiovisual stimuli were generated by changing the combinations of 12 sound stimuli and 3 visual stimuli.

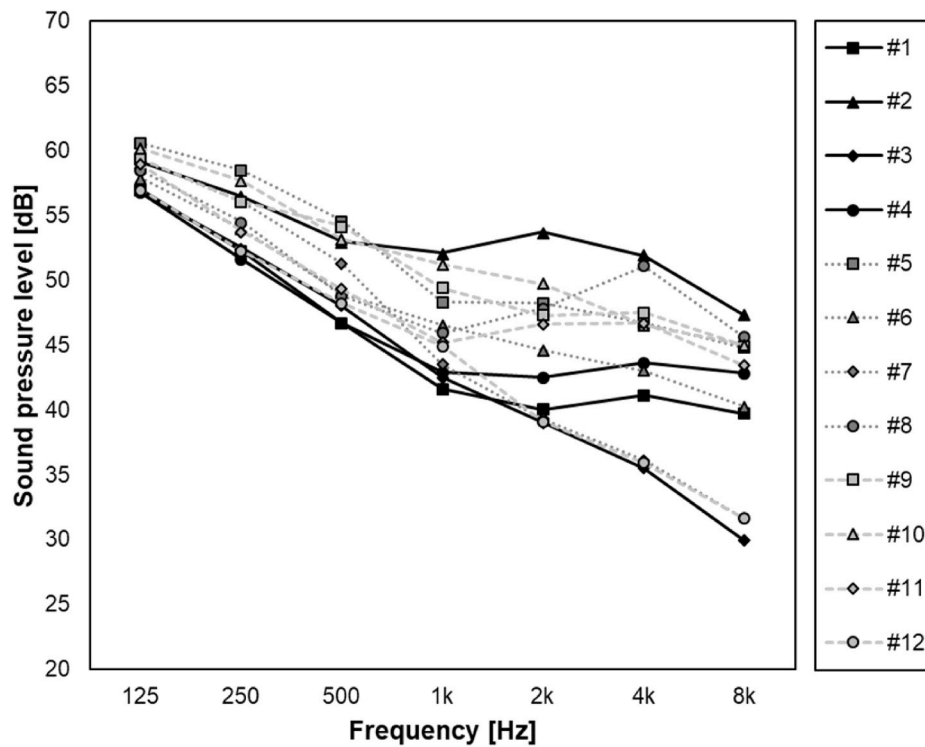
#### 3.1.2. Questionnaire

In Experiment 2, the questionnaire used in the subjective evaluation of the indoor soundscape of the OPO was composed of three parts. In the first part, noise source identification of various audio contents were examined. The participants were instructed to verbalize the degree of their perception of each sound stimulus from 12 noise sources. This

**Table 1**

Acoustic parameter values of sound stimuli.

| Acoustic parameter  | #1   | #2   | #3   | #4   | #5   | #6   | #7   | #8   | #9   | #10  | #11  | #12  |
|---------------------|------|------|------|------|------|------|------|------|------|------|------|------|
| $L_{Aeq}$ [dBA]     | 50.3 | 59.3 | 49.9 | 51.4 | 56.8 | 53.0 | 52.4 | 55.8 | 56.4 | 57.3 | 54.0 | 50.6 |
| $L_{A10-A90}$ [dBA] | 2.7  | 10.5 | 4.4  | 3.4  | 12.6 | 6.2  | 7.6  | 4.9  | 9.2  | 9.8  | 5.4  | 5.4  |

**Fig. 4.** 1/1 octave band frequency characteristics of the 12 sound stimuli.**Table 2**

Frequency of noise source occurrence.

| Noise source              | #1 | #2 | #3 | #4 | #5 | #6 | #7 | #8 | #9 | #10 | #11 | #12 |
|---------------------------|----|----|----|----|----|----|----|----|----|-----|-----|-----|
| Conversation [s]          | 3  | 4  | 12 | -  | 47 | 29 | 55 | 45 | 37 | 60  | 26  | 22  |
| Laughter                  | -  | -  | 2  | -  | -  | 1  | -  | -  | 3  | 5   | 2   | 8   |
| Keyboard and mouse        | 12 | 42 | 46 | 72 | 20 | 49 | 32 | 39 | 31 | 25  | 34  | 85  |
| Walking and footsteps     | 7  | 6  | -  | -  | 10 | -  | -  | 20 | 15 | 5   | 36  | 14  |
| Chair dragging            | -  | 5  | 3  | -  | 2  | 4  | 6  | -  | 14 | 14  | 9   | 10  |
| Desk noise                | 1  | 5  | 8  | 5  | -  | 16 | 9  | -  | 7  | 4   | 3   | 2   |
| Page turning              | -  | 2  | 3  | 2  | -  | -  | 2  | -  | -  | -   | 8   | -   |
| Printers and copiers      | 3  | -  | -  | 6  | -  | -  | -  | -  | -  | -   | -   | 3   |
| Doors slamming            | -  | 1  | 1  | -  | 3  | -  | -  | 2  | -  | 4   | -   | -   |
| Water purifier            | -  | 4  | -  | -  | 4  | -  | -  | -  | -  | -   | 2   | -   |
| Glass cup clinking        | -  | 3  | -  | -  | 3  | -  | -  | -  | -  | -   | -   | -   |
| Mechanical fan and AC [s] | 60 | 60 | 60 | 60 | 60 | 60 | 60 | 60 | 60 | 60  | 60  | 60  |
| Total noise event type    | 6  | 10 | 8  | 5  | 8  | 6  | 6  | 5  | 7  | 8   | 9   | 8   |

noise source classification system was designed to include all noise sources generated from the sound source samples listed in Table 2. In the second part, the participants were instructed to respond to the degree of matching with eight adjectives for each stimulus (full of content, interesting, comfort, private, empty, boring, annoying, and dynamic) to examine the perceived affective quality of the indoor soundscape. This adjective terminology system was selected with reference to eight adjective terms selected by Torresin et al. [45] to analyze the indoor environment based on the comfort–content dimension. This study presented 8 indoor soundscape evaluation measures composed of a 45° unit axis based on 97 adjective terms. In the final part, the overall work-related quality was divided into satisfaction, willingness to work,

and perceived productivity, and the participants were instructed to evaluate the degree of agreement with each stimulus. Furthermore, the participants were instructed to rate each metric described above based on a seven-point Likert-like scale (0 = *not at all* to 6 = *extremely*).

### 3.1.3. Procedure

A total of 41 participants (23 males and 18 females) participated in the study, and their ages ranged from 20 to 29 years old (mean age: 23.12 and standard deviation: 2.82). The subject group that participated in Experiment 2 was included as an independent group comprising subjects that did not participate in Experiment 1. College and graduate school students from the same university were selected to reduce the

variations in the participant responses. Only participants who had experience of working in an office space were recruited. The sample size was calculated using G-Power software (version 3.1.9.4). For verification with an effect size of 0.30 and power of 0.8, at least 36 participants were required. The criterion was met because a total of 41 participants were selected. Before starting the experiment, written consent regarding participation in this study was obtained from every subject as an ethical procedure. Furthermore, the purpose of the experiment and contents of the questionnaire were explained to the participants. Normal visual and auditory functions were verified for each subject through self-diagnosis. In addition, the use of VR devices was explained, and they were allowed to sufficiently familiarize themselves with the devices. The evaluation environment was constructed in a semi-anechoic room, where the background noise was very low at approximately 30 dBA. The temperature in the laboratory environment was maintained at 23–24 °C, which was suggested as an appropriate temperature for an office space in previous studies [63,64].

A total of 36 stimuli (12 audio stimuli  $\times$  3 visual stimuli) were randomly presented to the participants. The sound and visual stimuli were presented using the same equipment and methods used in Experiment 1. The subjects experienced each evaluation stimulus for 1 min before responding to the questionnaire. The stimuli were provided repeatedly according to the subjects' requests. The questionnaire was provided through a GUI in a VR environment. After the evaluation for one stimulus was completed, silence was maintained for 10 s before the next stimulus was presented, and this process was repeated. The total duration of the experiment was limited to less than an hour considering the fatigue of subjects and adverse reactions (e.g., dizziness, nausea) from wearing the VR equipment for a long time. A total of 1476 responses (36 stimuli  $\times$  41 participants = 1476) were collected for each evaluation item.

## 3.2. Results

### 3.2.1. Noise source identification

The degrees of identification of various noise sources generated in the OPOs are listed in Table 3. The analysis result of the Pearson correlation coefficient between the occurrence frequency and degree of identification of each noise source is summarized in Table 3. Conversation was the most prominent source among the various noise sources in the OPOs, and it exhibited a high positive correlation of 0.79 ( $p < 0.01$ ) with the actual occurrence frequency. The correlation with laughter was high at 0.81 ( $p < 0.01$ ), which implies the sound of laughter (#9 and #12) was well perceived by the participants. This high correlation between the actual frequency of speech-related noises in OPOs with the degree of identification indicated that workers were relatively sensitive to speech noise. Following speech-related noise, the identification of the sounds of a keyboard and mouse was dominant and exhibited a significant correlation of less than 0.44 with the occurrence frequency ( $p < 0.01$ ). The mechanical fan and AC had the longest

duration of over 60 s among all sources of sound stimuli; however, they were not accurately identified by the subjects. This was because the subjects reacted less sensitively to these noises as background noise instead of foreground noise. Furthermore, several noise sources such as glass cup clinking (#2), walking and footsteps (#8), and laughter (#9, #12) could be identified by most subjects. However, most occurrences of these stimuli were not accurately identified by the subjects when compared with the actual occurrence frequency.

### 3.2.2. Perceived affective quality and work-related quality

The average perceived affective quality responses for the indoor soundscape of OPOs are shown in Fig. 5. Here, eight adjective terms were marked at 45° intervals by referring to a previous study [65]. The perceived affective quality responses can be classified into two types:

- 1) Average responses for the private, comfortable, empty, and boring dimensions were high for the sound environments with low conversation noise due to speech (#1 and #4).
- 2) Perceptions of the dynamic, full of content, and annoying dimensions were higher when there were more types of noise contents in the office sound environment.

Even if the frequency of the speech-related noise was low, the private, comfortable, and empty dimensions decreased when the number of noise source types was high, whereas the dynamic, full of content, and annoying dimensions increased (#2). In contrast, even if the frequency of conversation was high, private and controlled noise did not decrease significantly when several other noise sources occurred. This can be attributed to the masking effect of other noise sources (#3).

Results for the responses for work-related quality according to the eight perceived affective qualities are illustrated in Fig. 6 (a)–(h). Linear regression analysis was performed to examine the correlations between the two evaluation factors. Regression analysis models other than that for the correlation between the interest and the perceived productivity were statistically significant. The correlation of satisfaction and willingness to work with perceived productivity was opposite to the correlation with perceived affective quality. The preferred zone (satisfaction and willingness to work  $>$  perceived productivity) and productive zone (satisfaction and willingness to work  $<$  perceived productivity) were defined based on these correlations. Here, the preferred zone is a space that reflects the need of the workers to work in a comfortable environment, whereas a productive zone is a space required by managers who aim to increase productivity.

First, the full of content, dynamic, and annoying dimensions for the indoor soundscape of the OPOs exhibited a strong correlation of 0.91 or higher with perceived productivity. Thus, with an increase in the identification of these dimensions, the space will become a more productive zone. In contrast, satisfaction and willingness to work exhibited a strong negative correlation of  $-0.83$  or higher with full of content, dynamic, and annoying. The lower the perceived affective quality, the more this

**Table 3**  
Mean rating score of noise source identification and Pearson correlation with frequency of noise occurrence.

| Noise source          | <i>r</i> | #1  | #2  | #3  | #4  | #5  | #6  | #7  | #8  | #9  | #10 | #11 | #12 |
|-----------------------|----------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Conversation          | 0.79**   | 0.9 | 1.9 | 1.6 | 0.3 | 4.8 | 3.8 | 4.2 | 1.3 | 4.3 | 4.5 | 2.0 | 1.8 |
| Laughter              | 0.81**   | 0.4 | 0.4 | 2.0 | 0.3 | 1.0 | 1.3 | 0.5 | 0.3 | 3.6 | 1.3 | 2.3 | 4.0 |
| Keyboard and mouse    | 0.44**   | 3.7 | 2.1 | 2.2 | 3.0 | 2.2 | 3.4 | 3.1 | 1.9 | 2.3 | 2.0 | 1.7 | 2.5 |
| Walking and footsteps | 0.58**   | 1.0 | 1.1 | 0.6 | 1.1 | 1.8 | 0.8 | 0.5 | 4.0 | 1.9 | 1.3 | 2.6 | 1.1 |
| Chair dragging        | 0.23     | 0.3 | 1.5 | 0.7 | 0.7 | 0.8 | 0.7 | 0.4 | 0.6 | 0.9 | 1.0 | 0.6 | 0.5 |
| Desk noise            | -0.13    | 0.9 | 2.9 | 1.3 | 1.7 | 2.0 | 1.4 | 0.9 | 1.7 | 1.6 | 2.6 | 1.4 | 1.0 |
| Page turning          | 0.59**   | 0.4 | 1.1 | 0.6 | 0.5 | 0.7 | 0.5 | 1.0 | 0.5 | 0.7 | 0.6 | 1.1 | 0.8 |
| Printers and copiers  | 0.15     | 1.4 | 1.4 | 0.6 | 1.2 | 1.1 | 1.1 | 0.8 | 1.5 | 1.5 | 1.2 | 1.4 | 1.3 |
| Doors slamming        | 0.38**   | 1.4 | 1.4 | 0.7 | 1.7 | 1.3 | 1.1 | 0.6 | 2.0 | 1.7 | 2.1 | 1.8 | 0.6 |
| Water purifier        | 0.58**   | 0.5 | 1.6 | 0.2 | 0.5 | 1.8 | 0.5 | 0.3 | 0.6 | 0.7 | 2.1 | 0.7 | 0.6 |
| Glass cup clinking    | 0.95**   | 0.4 | 3.9 | 0.3 | 0.5 | 2.6 | 0.4 | 0.3 | 0.6 | 0.9 | 0.7 | 0.5 | 0.3 |
| Mechanical fan and AC | -        | 1.3 | 1.3 | 1.2 | 1.5 | 1.2 | 1.3 | 1.2 | 1.3 | 1.4 | 1.2 | 1.4 | 1.1 |



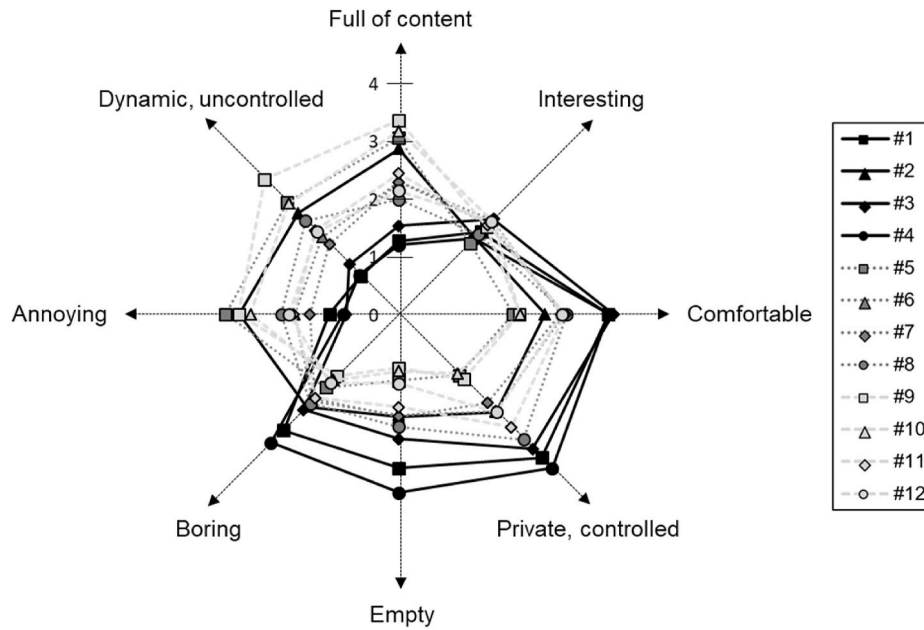


Fig. 5. Perceived affective quality of indoor soundscape in open-plan offices.

space is considered as a preferred zone. The comfortable and private dimensions exhibited a significantly strong negative correlation with perceived productivity and a highly positive correlation of 0.83 with satisfaction and willingness to work. Furthermore, the empty and boring dimensions exhibited a similar trend as that of the comfortable and private dimensions; this indicates a distinct correlation of  $-0.68$  or higher with perceived productivity and  $0.58$  or higher with satisfaction and willingness to work. Finally, the interesting dimension exhibited weak correlations of  $0.34$  and  $0.39$  with satisfaction and willingness to work, respectively, when compared with other perceived affective qualities.

In summary, the sound environment should be designed in the descending order of importance of the following dimensions to achieve a preferred zone from the standpoint of the workers: comfortable, private, empty, boring, and interesting. The office sound environment should be designed in the descending order of importance of the following dimensions to achieve a productive zone from the perspective of managers: full of content, annoying, and dynamic. In the final analysis, comfortable–annoying, full of content–empty, and private–dynamic were found to be important dimensions among the four different dimensions shown in Fig. 5 for analyzing the indoor soundscape of OPOs in terms of the work-related quality.

For a more intuitive analysis of the evaluation results for the eight adjective terms, the perceptual dimension of the indoor soundscape was reduced to the comfort–content axis, and the factor score for each dimension was respectively calculated using Equations (1) and (2). For this factor score calculation, the responses to eight adjective terms in Fig. 5 were determined through orthographic projections on the comfort and content axes by referring to a previous study [65]. This factor score calculation method was consistent with the method of calculating the coordinate score for the perceptual dimension in the soundscape evaluation for external spaces in ISO 12913-3 [39]. The private–dynamic axis was excluded because the former can be interpreted as a region with high comfort and low content, and the latter can be interpreted as a region with low comfort and high content. Thus, only the two comfort–content axes were distinguished.

$$\text{Comfort} = (co - an) + \cos 45^\circ \cdot (pi - dy) + \cos 45^\circ \cdot (in - bo), \quad (1)$$

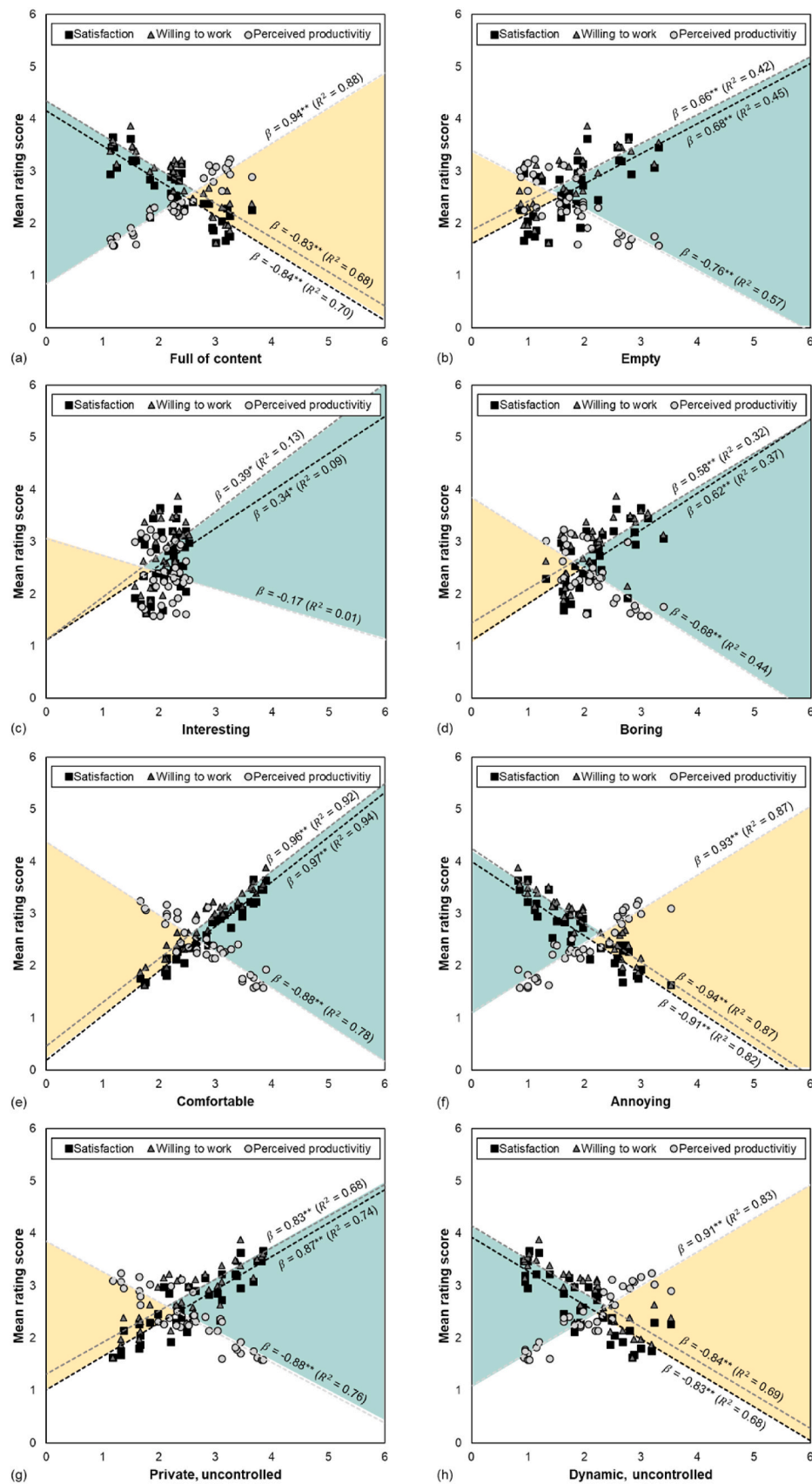
$$\text{Content} = (fu - em) + \cos 45^\circ \cdot (dy - pi) + \cos 45^\circ \cdot (in - bo), \quad (2)$$

where *co*, *an*, *pi*, *dy*, *in*, *bo*, *fu*, and *em* represent comfort, annoying, private, dynamic, interesting, boring, full of content, and empty, respectively.

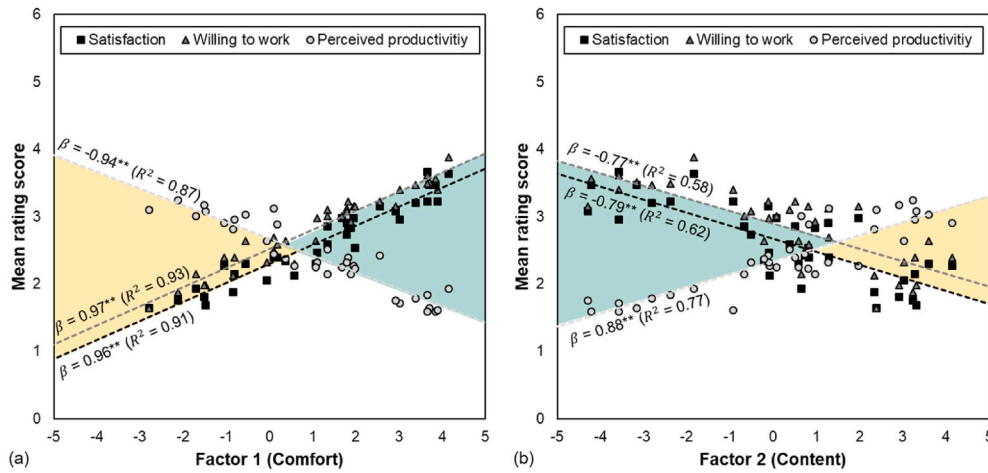
The relationships between comfort and content and work-related quality are shown in Fig. 7 (a) and (b). First, for the indoor soundscape of the OPO, a higher comfort and lower content are more advantageous for an office space to be considered a preferred zone. In contrast, the office space should have a lower comfort and a higher content to be considered a productive zone. Based on the factor score of the comfort–content dimensions, when comfort was in the range of  $0.28$ – $0.68$  and content was in the range of  $0.58$ – $1.16$ , the office space corresponded to a balanced section between the preferred and productive zones. Based on the median value of the score, which was zero, the content of the indoor soundscape in the office should be at an average level or higher. The average response for willingness to work was higher than the average response for satisfaction by approximately  $0.2$  or more. This suggests that even if worker satisfaction with the OPO environment does not meet their expectations, they have a higher willingness to participate in work.

### 3.2.3. Perceptual indoor soundscape model in OPO

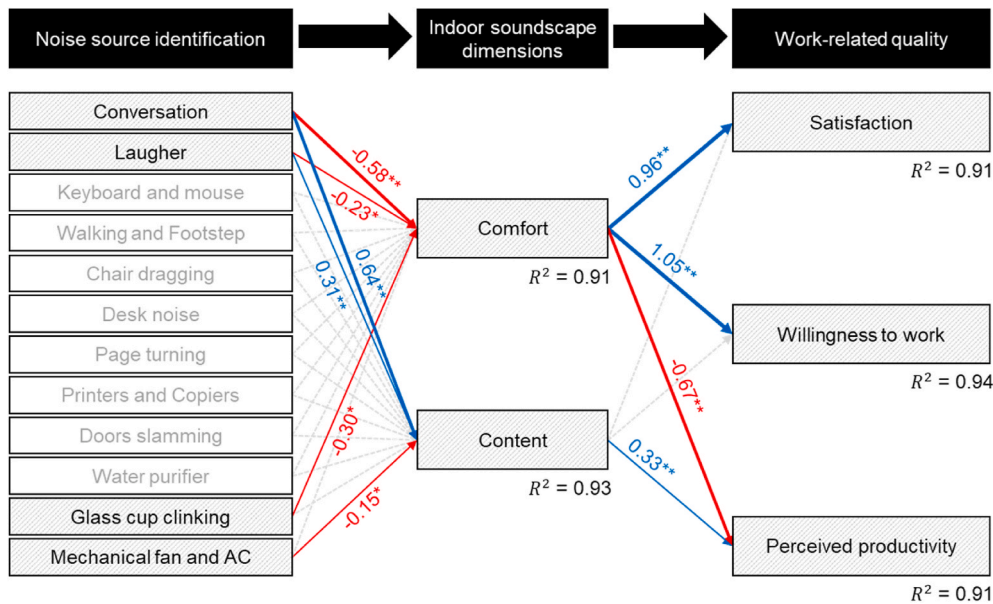
A perceptual indoor soundscape model was proposed by conducting a linear regression analysis between the evaluation factors to comprehensively investigate the correlations among noise source identification, perceived affective quality, and work-related quality; the results are shown in Fig. 8. Overall,  $R^2$  was  $0.91$  or higher, which indicates that the model is statistically significant. First, the examination of the correlations between noise source identification and indoor soundscape dimensions reveals that conversation and laughter among workers have negative effects of  $-0.58$  and  $-0.23$  on the comfort, respectively; and positive effects of  $0.64$  and  $0.31$  on the content, respectively. Furthermore, fan noise, which was generated continuously by the HVAC system in the OPOs, did not have a significant effect on comfort but had a negative effect of  $-0.15$  on content. The clinking sounds of glass cups used for drinking by workers during work or meetings had a negative effect of  $-0.30$  on comfort. When the correlations between the indoor soundscape dimensions and work-related quality were examined, comfort had positive effects of  $0.96$  and  $1.05$  on satisfaction and willingness to work, respectively, and a negative effect of  $-0.67$  on perceived productivity. These results are in good agreement with the trends observed



**Fig. 6.** Relationship between perceived affective quality and overall quality: (a) full of content, (b) empty, (c) interesting, (d) boring, (e) comfortable, (f) annoying, (g) private, and (h) dynamic. Green: preferred zone, yellow: productive zone. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)



**Fig. 7.** Relationship between indoor soundscape perceptual dimension and overall preference. (a) Comfort and (b) content. The green color indicates the preferred zone, and the yellow color indicates the productive zone. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)



**Fig. 8.** Indoor soundscape perception model in OPO (the numbers indicate the standardized regression coefficients ( $\beta$ )).

in Fig. 7. Furthermore, content had a positive effect of 0.33 on the perceived productivity.

Finally, the correlations between the indoor soundscape perception factors and physical acoustic parameters are presented in Table 4. First,  $L_{Aeq}$  exhibited statistically significant correlations with comfort and perceived productivity, whereas  $L_{A10-A90}$  exhibited statistically significant and stronger correlations with all parameters than  $L_{Aeq}$  compared

**Table 4**

Standardized regression coefficient ( $\beta$ ) with physical acoustic parameters and subjective response.

| Descriptors                 | $R^2$ | $L_{Aeq}$ [dBA] | $L_{A10-A90}$ [dBA] |
|-----------------------------|-------|-----------------|---------------------|
| Indoor soundscape dimension |       |                 |                     |
| Comfort                     | 0.79  | $-0.35^{*}$     | $-0.59^{**}$        |
| Content                     | 0.64  | 0.04            | $0.78^{**}$         |
| Work-related quality        |       |                 |                     |
| Satisfaction                | 0.70  | $-0.20$         | $-0.68^{**}$        |
| Willing to work             | 0.69  | $-0.27$         | $-0.61^{**}$        |
| Perceived productivity      | 0.86  | $0.35^{**}$     | $0.62^{**}$         |

with all the parameters. This suggests that the temporal variation characteristic of the physical sound environment is a critical parameter for the evaluation of the indoor soundscape of OPOs. Second,  $L_{Aeq}$  had a negative effect of  $-0.35$  on comfort but a positive effect of  $0.35$  on perceived productivity. In addition,  $L_{A10-A90}$  had a negative effect of  $-0.59$  on comfort and a positive effect of  $0.78$  on content. When the correlation with work-related quality was examined,  $L_{A10-A90}$  exhibited negative effects of  $-0.68$  and  $-0.61$  on the satisfaction and willingness to work, respectively, and a positive effect of  $0.62$  on the perceived productivity. This was in good agreement with the increase in work performance in accordance with an increase in the acoustic variability beyond a certain level in Experiment 1 and with the increase in perceived productivity in accordance with an increase in the dynamics of the sound environment in Experiment 2. In summary, comfort decreased when multiple noise events occurred, the noise level was high, and the variability over time was significant in the OPOs. However, such noise variability can contribute to increased workability and the identification of content.

## 4. Discussions

### 4.1. Design strategy of OPO according to indoor soundscape perception model

Based on the two experiments and the indoor soundscape perception model of OPOs, a design strategy for the OPO sound environment to achieve preferences (satisfaction and willingness to work) and work-related quality was proposed in terms of the audiovisual content. Interestingly, satisfaction and willingness to work showed an opposite trend to that of perceived productivity with regard to the indoor soundscape quality. In some previous studies [6,66,67] as well, changes in the layout of the office space had opposite effects on satisfaction and performance. Furthermore, noise in an office space had a negative effect on satisfaction, but no clear correlation with work performance was found, which agreed with this finding [6,67]. Therefore, this study presented separate design strategies regarding the preference for the OPO environment and work performance. The recommended values of the physical acoustic parameters for the OPO sound environment are summarized in the ISO 22955 and ISO 3382-3 standards [10,28]. Therefore, this study proposes a design strategy for indoor soundscape perception based on the research questions considered in this study.

Q1) How does work-related quality change in accordance with changes in the audiovisual content?

As demonstrated in Experiment 1, the visual complexity levels required to achieve the maximum psychological satisfaction required by workers and the maximum physical work performance required by managers were  $-1.28$  and  $-0.10$ , respectively, with respect to visual complexity, which exhibits a difference of approximately  $1.18$ . The visual complexity should be decreased by providing a large internal layout and well-organized visual OPO space such as Office 4 to enhance satisfaction. However, an excessive decrease in the visual complexity of the internal space can lead to decreased work performance. Therefore, the OPO space should be designed to achieve an appropriate visual complexity range of approximately  $-1.28$  to  $-0.10$ . Furthermore, a design that decreases the acoustic variability to a maximum of  $-0.28$  is required, which can be achieved by decreasing the  $L_{p,A,S,4m}$  of the sound environment based on the sound absorption design of the OPO, to simultaneously enhance the satisfaction and work performance of workers in terms of acoustic variability.

Q2, Q3) What causes the correlations between the indoor soundscape perceptual dimension and work-related quality? How should the OPO indoor soundscape be designed to improve work performance and preference for the work environment?

Comfort should be increased to improve the preference for the OPO, and comfort should be decreased and content should be increased to improve the perceived productivity. Furthermore, to achieve a balance between the preferred and productive zones, it is appropriate to achieve comfort and content within the ranges of  $0.28$ – $0.68$  and  $0.58$ – $1.16$ , respectively. It is necessary to lower the identification of conversation or laughter by decreasing speech intelligibility in the office and achieving speech privacy to increase comfort. This can be implemented using various methods such as sound absorption design for the office interior (such as ceiling, floors, and walls), the installation of sound-absorbing objects or partition walls, and implementation of layout changes [10]. The most important element is the surface material, given that the high sound-absorbing power of materials increases the sound-absorbing power of the interior space, which leads to the blocking of strong reflections. In contrast, an opposite strategy is required to improve content. However, it is extremely difficult for designers to implement these two strategies simultaneously. Preference and perceived productivity can be realized simultaneously by providing content for workers participating in conversation, while improving comfort by securing privacy for other workers through a separate space module where tasks involving conversation occur, i.e., a system booth in the office. An alternative method is realizing a low-noise environment through a

separate duct sound absorption design to reduce mechanical noise or using a floor AC instead of a ceiling AC.

Finally, with respect to the physical acoustic parameters, which are different from the decrease in overall work performance with an increase in  $L_{p,A,S,4m}$ , as observed in Experiment 1, the perceived productivity increased with an increase in  $L_{Aeq}$ . This was because there was no significant difference in the acoustic contents in Experiment 1, given that the same speech sound source was targeted. However, in Experiment 2, wherein sound sources with different acoustic contents were targeted,  $L_{A10-A90}$  varied significantly even when the  $L_{Aeq}$  values were similar. This difference gave rise to the different relationships of  $L_{Aeq}$ . Therefore, it is necessary to maintain a low  $L_{Aeq}$  of the background noise inside the OPO while increasing  $L_{A10-A90}$  by diversifying the acoustic contents.

### 4.2. Limitations and scope of future research

This study had several limitations. First, the participants comprised a small group of college and graduate school students (age range: 20–29 years old), and the male-to-female ratio was not balanced. Thus, this study did not consider the entire working age group, and the proposed generalized perception model was limited because of the limited audiovisual office stimuli used in the experiments. Furthermore, compared with the actual OPO context, the sense of reality was somewhat reduced because of the absence of senses other than the audiovisual senses, given that short-term reactions to the audiovisual environment were obtained using the VR technology in a limited laboratory environment.

Another limitation is that long-term effects in actual offices were not examined because worker responses to short-term stimuli in a laboratory environment were considered. Future research should involve the examination of the correlations between the perception of the indoor soundscape and various physical acoustic parameters suggested in the ISO 22955 or ISO 3382-3 standards [10,28] in addition to examining  $L_{Aeq}$  and  $L_{A10-A90}$ . If our research is extended to various office spaces, it is expected that the effects of room acoustic parameters such as reverberation time and early decay time on the work-related quality in terms of the indoor soundscape can be verified. Lastly, we will be able to closely examine the soundscape recognition system for each worker by considering various socio-demographic characteristics such as the type of living space and the level of education of the subjects [68]. However, despite such limitations, this study is significant because it provides a new perspective on the need to secure the content of a sound environment with an appropriate level of noise events with respect to the indoor soundscape, which differs from the conventional design strategy that assumes quietness as an overall advantage for OPOs.

## 5. Conclusions

This study proposed a perception model to perform subjective evaluations in a VR environment considering audiovisual senses by investigating the effects of various audiovisual contents in OPOs on indoor soundscape perception. An indoor soundscape design strategy for OPOs was developed based on the proposed model. The findings of this study are as follows.

- 1) Different strategies are required to ensure the satisfaction and work performance of workers because they exhibit opposite trends in terms of the indoor soundscape. A relatively low acoustic variability and visual complexity are required to achieve worker satisfaction and work performance.
- 2) It is critical to provide comfort and content to increase worker satisfaction and work productivity. Given that comfort has a negative effect on the work productivity, it is necessary to maintain a good balance between the preferred and productive zones.

The findings of this study can be used as fundamental data for the



indoor soundscape design guidelines of OPOs, and are expected to be useful for managers who need to consider the satisfaction and productivity of workers simultaneously.

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## CRediT authorship contribution statement

**Hyun In Jo:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jin Yong Jeon:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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